iLearn Practice Exam

11BR · Chapter / Topic 3

Cambridge AS Level Computer Science 9618

Paper 1 Theory Fundamentals

Topic 3: Hardware

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: 75

Marks obtained: ..... / 75

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

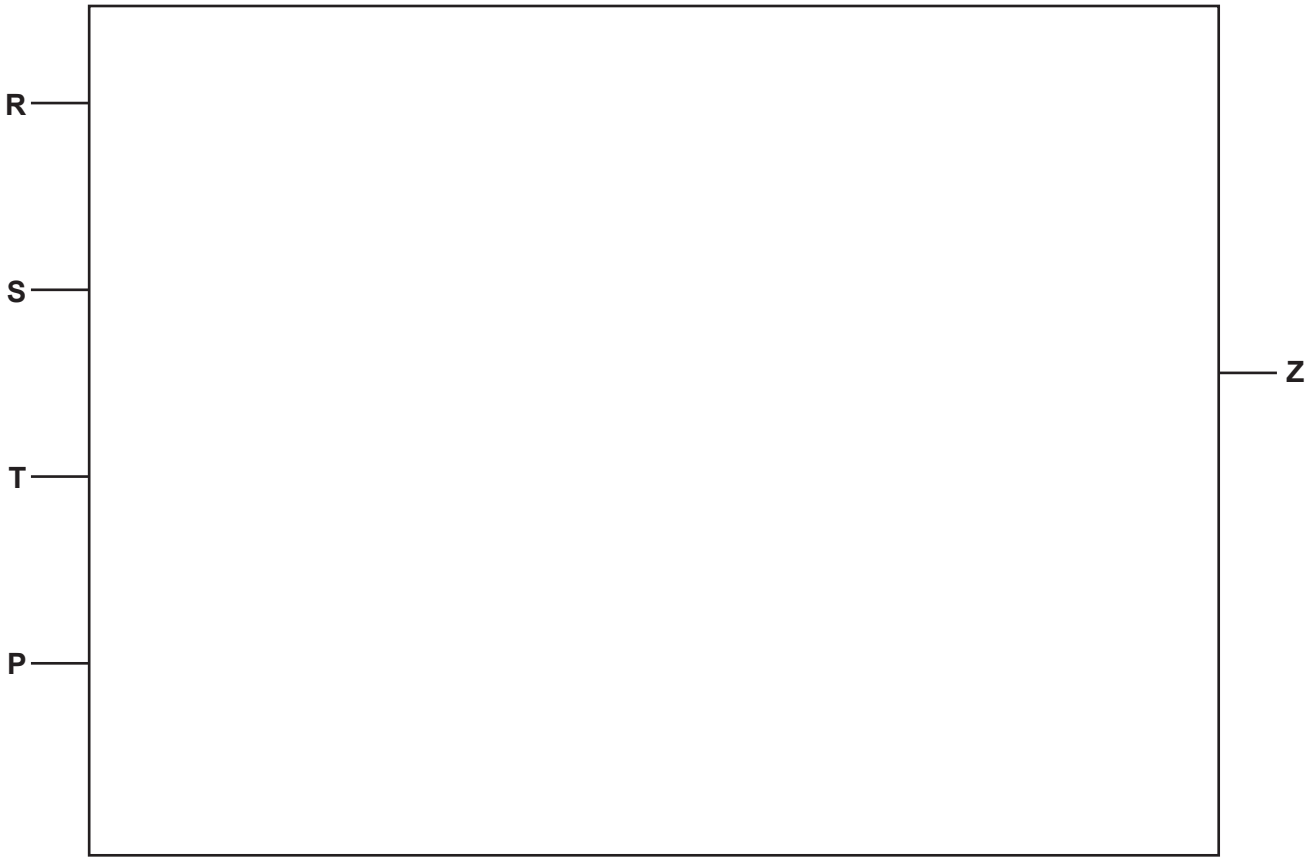
INFORMATION

· The total mark for this paper is 75.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 13 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 **(a)** Draw the logic circuit for this logic expression:

Z = (**R** XOR **S**) AND (NOT **T** NOR **P**)



[2]

(b) Complete the truth table for this logic expression:

Z = (NOT **P** OR **Q**) XOR (**R** NOR **Q**)

P	Q	R	Working space	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

2 **(a)** Describe the operation of each of the following logic gates:

NAND .....

.....

NOR .....

.....

XOR .....

.....

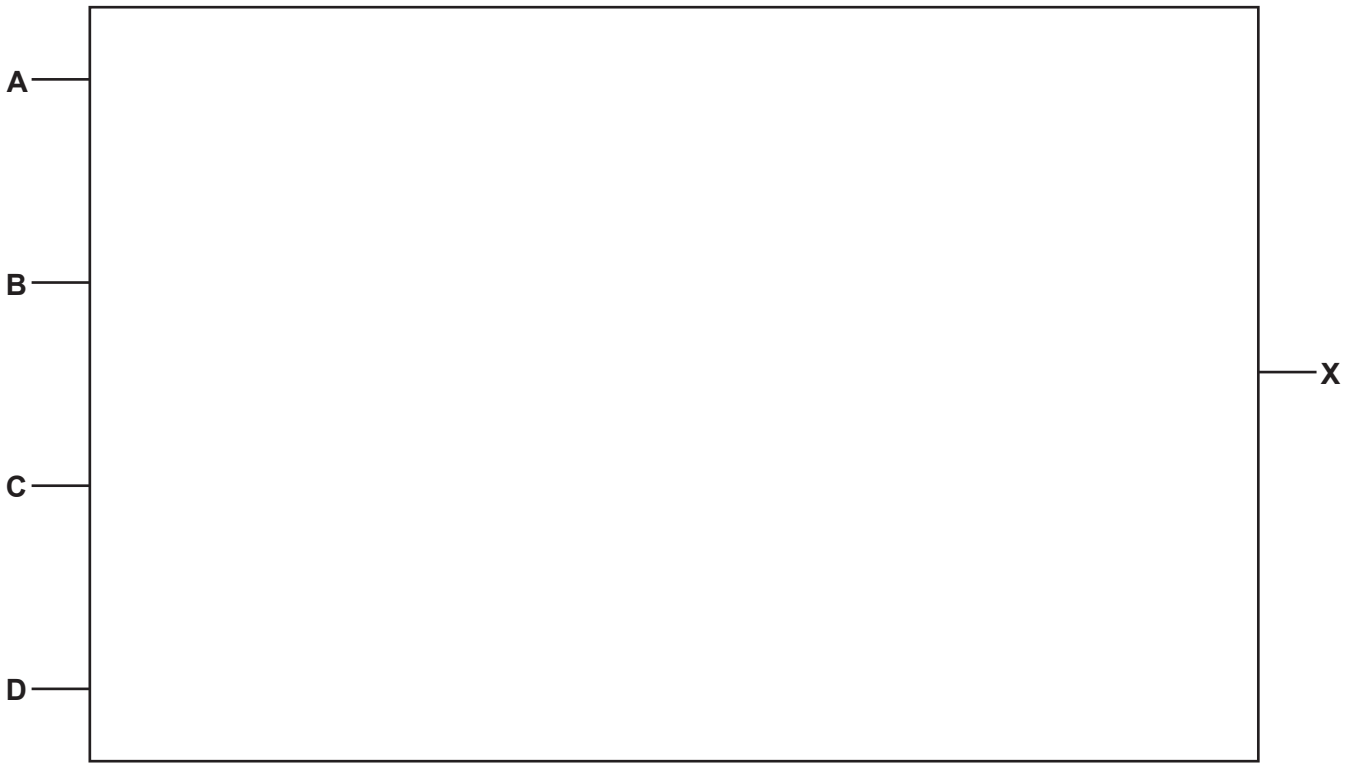
OR .....

.....

[4]

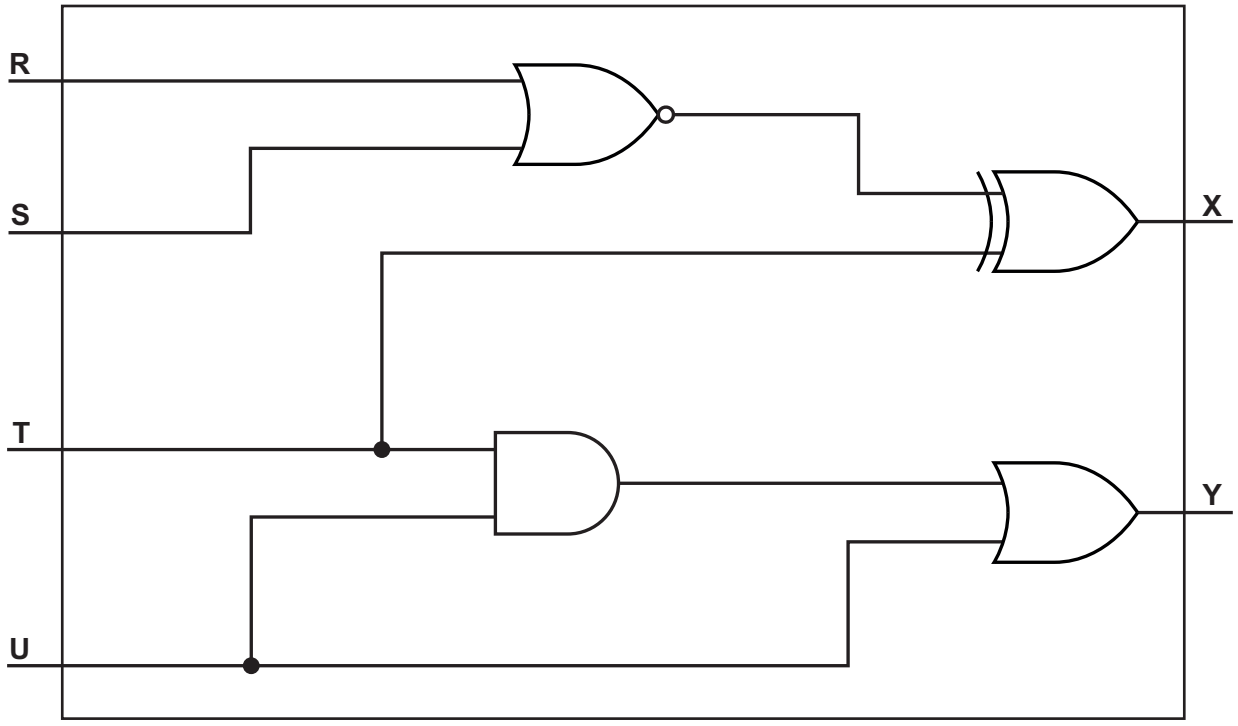
(b) Draw a logic circuit for this logic expression:

X = NOT ((**A** AND **B**) OR (**C** AND **D**))



[2]

3 **(a)** Write the logic expressions for the following logic circuit.



X = ...

Y = ...

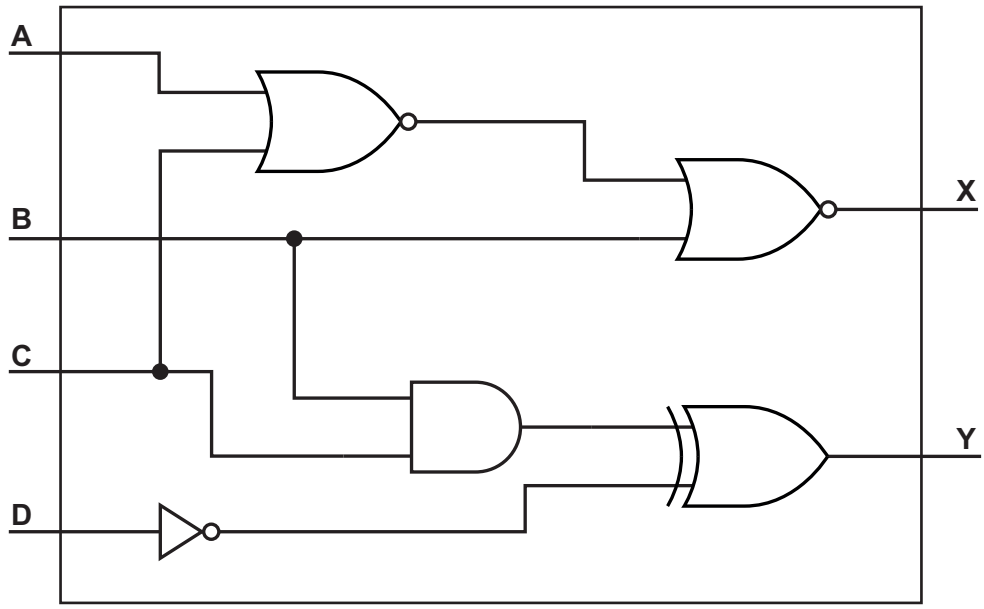
(b) Complete the truth table for the logic expression:

X = NOT (**A** AND (**B** XOR **C**))

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

4 **(a)** Write the logic expressions for the following logic circuit.



X = ...

Y = ...

[2]

(b) Complete the truth table for the logic expression:

X = (**A** OR **B**) XOR (**B** OR **C**) XOR (NOT **A** NAND **C**)

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

5 A computer is used to record sound.

(a) The computer has Programmable Read Only Memory (PROM).

Complete the table by naming and describing two other types of ROM.

type of ROM	description
	....
	....

[4]

(b) The sound is recorded using a microphone.

Describe the principal operation of a microphone.

.... [3]

(c) The computer outputs the sound through external speakers. The computer transfers the data to a buffer before the speakers output the sound.

Describe the purpose of the buffer.

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.....

.....

..... [3]

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6 A factory makes chocolate bars.

The factory uses a conveyor belt that moves the products from one stage of production to the next stage.

(a) An automated system counts the number of chocolate bars made at the end of production.

The system includes a sensor positioned above the conveyor belt.

Identify **one** appropriate type of sensor that can be used.

..... [1]

(b) A second automated system removes chocolate bars with an incorrect weight from the production line.

Describe the role of an **actuator** in this second system.

.....

.....

.....

..... [2]

(c) The factory has many different machines with embedded systems.

(i) Identify **two** features of embedded systems.

1 .....

.....

2 .....

.....

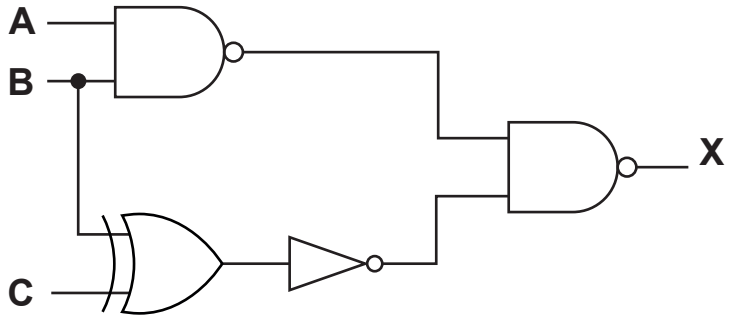
[2]

(ii) Identify **one** drawback of embedded systems.

.....

..... [1]

7 **(a)** Write the Boolean expression that corresponds to the following logic circuit.



.....

..... [3]

(b) Complete the truth table for the logic expression:

X = **A** XOR (**B** AND (**A** NAND **B**) AND NOT **C**)

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

8 **(a)** Complete the truth table for the logic expression:

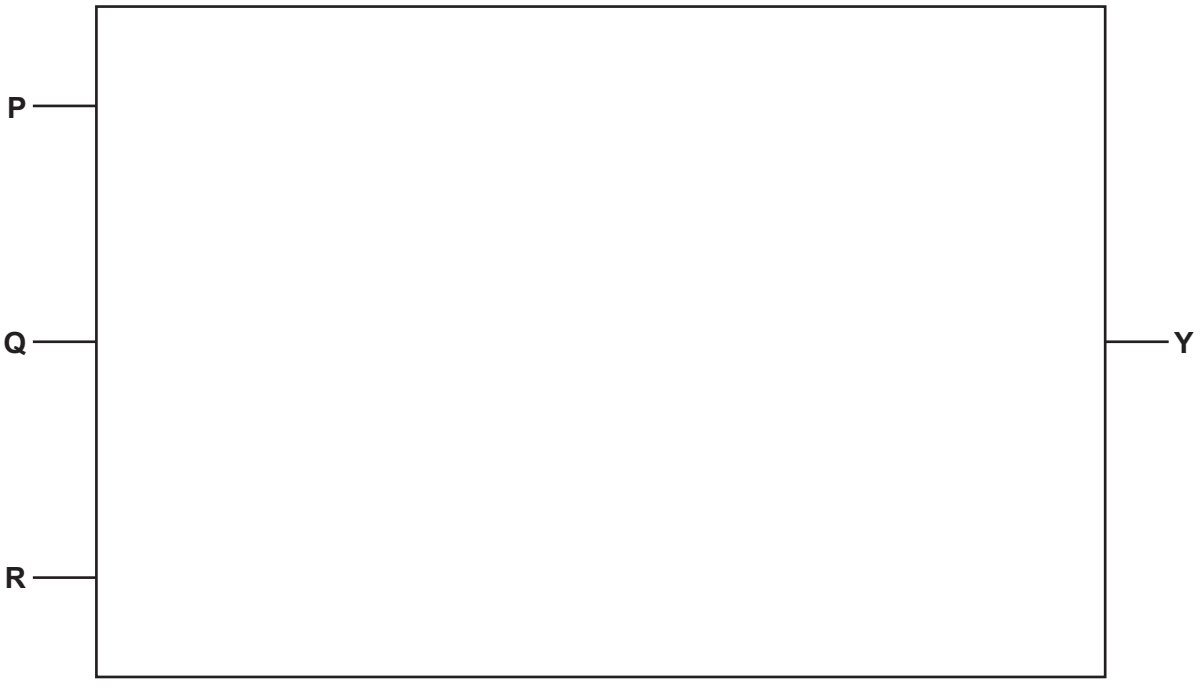
Y = ((**P** AND **Q**) XOR ((NOT **Q**) OR **R**)) AND NOT **P**

P	Q	R	Working space	Y
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(b) Draw a logic circuit for the logic expression:

Y = ((**P** AND **Q**) XOR ((NOT **Q**) OR **R**)) AND NOT **P**



[2]

9 A laptop computer has Static RAM (SRAM).

A virtual reality headset and a laser printer are connected to the laptop.

(a) Explain why Static RAM is used in the laptop instead of Dynamic RAM.

.....

.....

.....

..... [2]

(b) Identify **two** reasons for using Electrically Erasable Programmable ROM (EEPROM) in a virtual reality headset.

1 .....

.....

2 .....

.....

[2]

(c) Describe how the laser printer makes use of a buffer.

.....

.....

.....

.....

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.....

.....

..... [4]

(d) Identify **one** port that could be used to connect the virtual reality headset to the laptop.

Justify your choice.

Port .....

Justification .....

.....

.....

.....

[3]

10 Embedded systems are used in many electronic devices.

(a) Describe the drawbacks of embedded systems.

.....

.....

.....

.....

.....

..... [3]

(b) An embedded system has Dynamic RAM (DRAM).

Identify **one** benefit of using DRAM instead of Static RAM (SRAM) in an embedded system.

.....

..... [1]

(c) Give **two** differences between Erasable Programmable ROM (EPROM) and Electrically Erasable Programmable ROM (EEPROM).

Difference 1 .....

.....

.....

Difference 2 .....

.....

.....

[2]

11 A road bridge has a weight limit and a height limit for vehicles. For example, a vehicle must weigh less than 10 000 kg and must have a height of less than 3 m.

The bridge has a warning system. If a vehicle is approaching the bridge and it exceeds one or both limits, a sign displays a warning telling the driver of the vehicle to stop.

(a) The bridge warning system uses sensors to detect if a vehicle exceeds the limits.

Complete the table by identifying **two** different sensors that could be used by the system **and** describe how each sensor is used by the system.

Sensor	Use in bridge warning system
....	....
....	....

[2]

(b) Explain whether the bridge warning system is an example of a monitoring system or of a control system.

.....

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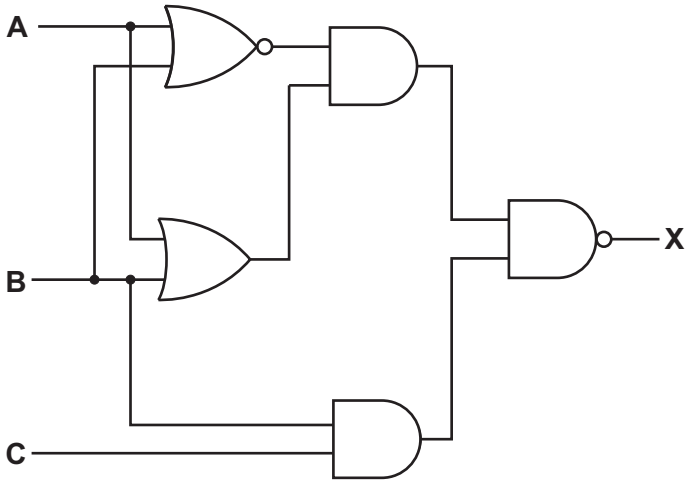
..... [2]

12 **(a)** Describe the operation of a 2-input XOR gate.

.....

..... [1]

(b) Write the logic expression for the following logic circuit.

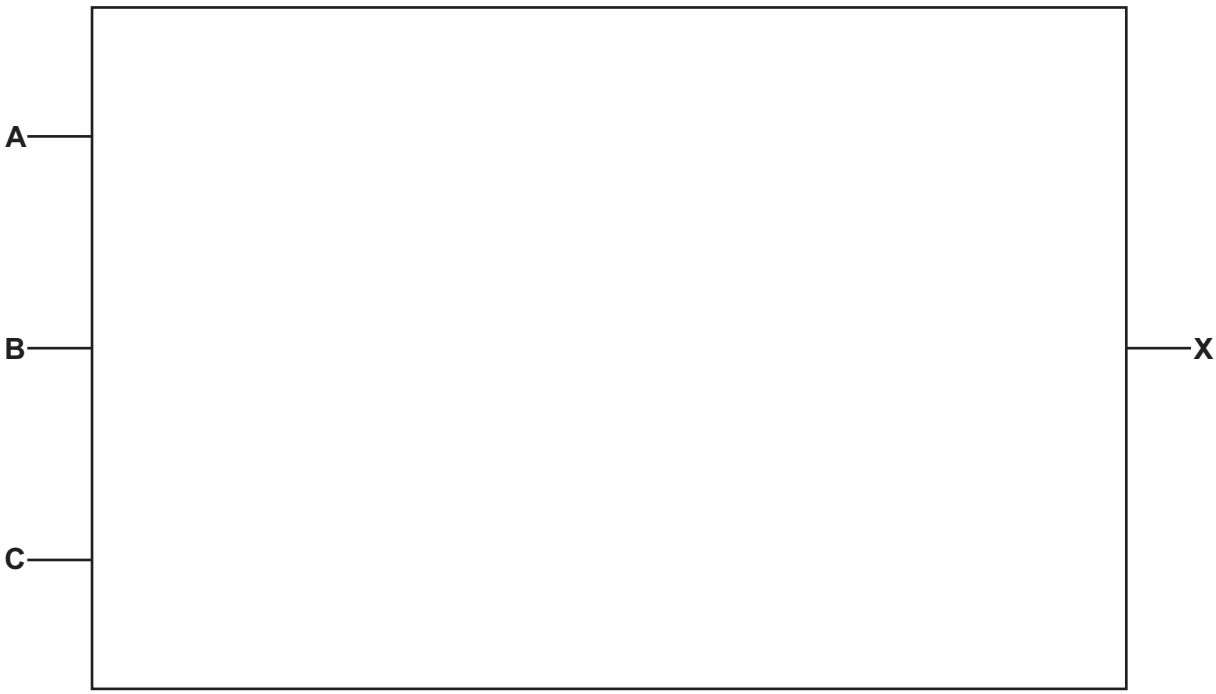


X = ...

..... [2]

(c) Draw the logic circuit for the logic expression:

X = ((NOT **A** AND (**B** AND **C**)) OR (**B** NAND **C**)) AND NOT **A**



[2]

13 A memory buffer uses Dynamic RAM (DRAM).

(a) Identify **two** differences between DRAM and Static RAM (SRAM).

Difference 1 .....

.....

.....

Difference 2 .....

.....

.....

[2]

(b) Explain how a memory buffer is used when a computer is transferring data to its magnetic hard disk drive.

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..... [4]

END OF QUESTION PAPER